

HAT-P-37b

R-2581

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_150813 · created 2026-08-27T00:52:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457247.8739 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.047
Transit depth (Rp/Rs)^2	2.38 +/- 1.45 [%]
Orbital Inclination (inc)	83.5 +/- 2.22
Airmass coefficient 1 (a1)	1.27 +/- 0.025
Airmass coefficient 2 (a2)	-0.123 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	1.97 %
Best Comparison Star	None
Optimal Aperture	3.53
Optimal Annulus	7.32
Transit Duration (day)	0.047 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.047 vs 0.1394 (deviation 0.29σ)
Duration fit vs published	0.047 d vs 0.0983 d (deviation 1.41σ)
Mid-time O–C	-2.2 min
Depth SNR	3.1
Residual scatter	1.97 %

Light curve

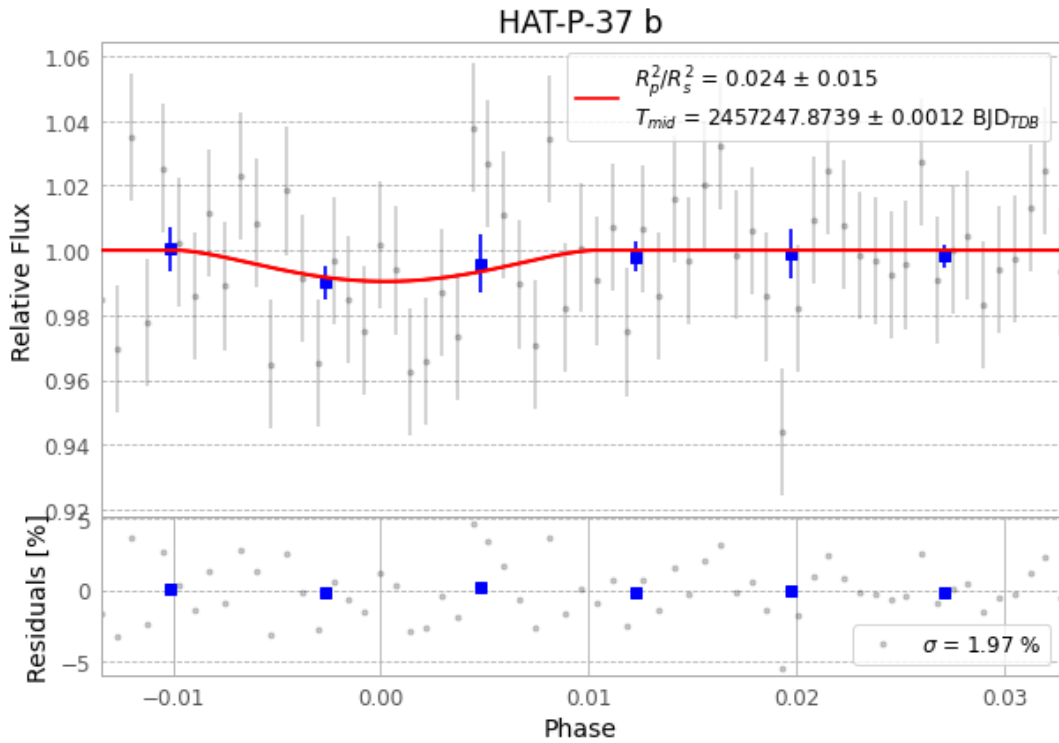


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-08-13.png (SHA-256 ef870f220e42de26...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 982395cbbd4773fc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), HATP-37150813080012.FITS → HATP-37150813110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150813.log (2db22b6bb9df...), FinalLightCurve_HAT-P-37 b_2015-08-13.png (ef870f220e42...), FinalLightCurve_HAT-P-37 b_2015-08-13.csv (23d9efdf08cc...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:52Z.